Using Potential Variables in Studies of Genetic Effects

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Motivation

One of the critical issues in genetic association studies is to evaluate association between genetic risk and human complex diseases.

$$\text{logit Pr}(Y = 1|X, G) = \alpha + \gamma X + \beta_1 G_1 + \cdots + \beta_m G_m$$

To test

$$H_0 : \beta_1 = \cdots = \beta_m = 0 \quad \text{vs} \quad H_1 : \exists \beta_i \neq 0 (i = 1, \ldots, m)$$

Motivation

Single Marker Analysis

Traditional Methods: test statistics derived under a specified genetic model by assigning a particular set of scores to the genotypes (Score Test, Wald Test, Likelihood Ratio Test)

Robust Methods: test statistics derived under the unknown genetic model (Pearson's Chi-Squared Test, MERT, MAX, CLRT, MAX3, MIN2, MET)

Motivation

To realize a better comprehending of biologic mechanisms for the susceptibility loci, it is critical to evaluate the risk of diseases associated with gene-gene or gene-environment interactions.

$$\text{logit Pr}(Y = 1|X, G) = \alpha + \gamma X + \beta_1 G_1 + \beta_2 G_2 + \beta_3 G_1 * G_2$$

$$\text{logit Pr}(Y = 1|X, E, G) = \alpha + \gamma X + \beta_1 E + \beta_2 G + \beta_3 G * E$$

Motivation

Existing Methods

- Case-Control analysis
- Case-Only analysis (Piegorsch et al. (1994))
- Semiparametric framework for retrospective maximum-likelihood analysis (Chatterjee and Carroll (2005))
- Empirical Bayes-type shrinkage estimator (Mukherjee et al. (2008))

Motivation

Score test derived under the additive genetic model (AST) was widely employed, which assigning a particular set of scores to the genotypes.

In practice, the underlying genetic modes of inheritance are rarely known, and miss-specifying a genetic mode may result in infertile tests.

Penetrance and Genotype Relative Risks

Consider a biallelic SNP locus with two alleles A and a, without loss of generality, let A be the risk allele.

Let the three genotypes be $G_0 = aa$, $G_1 = Aa$ and $G_2 = AA$. The allele frequencies in the population are denoted by $p = Pr(A)$ and $q = Pr(a) = 1 - p$. The population genotype probabilities of G_i are denoted by $g_i, i = 0, 1, 2$.

If Hardy-Weinberg proportions hold in the population,

$$g_0 = q^2, \ g_1 = 2pq, \ and \ g_2 = p^2$$

Penetrance and Genotype Relative Risks

The disease prevalence in the population is denoted by $k = \Pr(case)$. The penetrance of a disease given a genotype is denoted by $f_i = \Pr(case|g_i), i = 0, 1, 2$.

$$k = \sum_{i=0}^{2} Pr(g_i)Pr(case|g_i) = f_0 g_0 + f_1 g_1 + f_2 g_2.$$

Denote Genotype Relative Risks(GRR) by $\lambda_i = f_i / f_0, i = 1, 2$, where $f_0 > 0$.

Genetic Models

A genetic model refers to a specific mode of inheritance.

- the recessive model(REC)
- the additive model(ADD)
- the multiplicative model(MUL)
- the dominant model(DOM)

Genetic Models

Assume that A is a high risk candidate allele and a is any of the other alleles. Then

$$f_2 \geq f_1 \geq f_0 \ \textit{and} \ f_2 > f_0$$

- $f_1 = f_0$ or $\lambda_1 = 1$ (REC)
- $f_1 = (f_0 + f_2)/2$ or $\lambda_1 = (1 + \lambda_2)/2$ (ADD)
- $f_1 = \sqrt{f_0 f_2}$ or $\lambda_1 = (\lambda_2)^{1/2}$ (MUL)
- $f_1 = f_2$ or $\lambda_1 = \lambda_2$ (DOM)

The space Λ defines a family of genetic models indexed by $\theta \in [0,1]$ as

$$\Lambda = \{(\lambda_1, \lambda_2) : \lambda_1 = (1-\theta) + \theta\lambda_2, 0 \leq \theta \leq 1\}.$$

Models

We mainly use the following two logistic regression models for testing gene-gene and gene-environment interactions, respectively.

Model I

$$\text{logit Pr}(Y_k = 1) = \beta_0 + \beta_1 X_{1k} + \beta_2 X_{2k} + \beta_3 G_{1k} + \beta_4 G_{2k} + \beta_5 G_{1k} * G_{2k}$$

Model II

$$\text{logit Pr}(Y_k = 1) = \beta_0 + \beta_1 X_{1k} + \beta_2 X_{2k} + \beta_3 E_k + \beta_4 G_{1k} + \beta_5 G_{1k} * E_k$$

Models

Assigning a set of scores $s = (s_0, s_1, s_2)$ to genotypes (aa, Aa, AA). Followed by Sasieni et al.(1997), $s = (0, 0.5, 1)$ is assigned for the additive model, $s = (0, 1, 1)$ for the dominant model, and $s = (0, 0, 1)$ for the recessive model.

For many complex diseases, their underlying genetic models are often uncertain, that there is no single efficiency test for all scenarios.

The genotypes are coded by $(0, \theta, 1), \theta \in [0, 1]$, where θ can be regarded as a model parameter.

Models

Model I′

$$\begin{aligned}\text{logit Pr}(Y_k = 1) =& \beta_0 + \beta_1 X_{1k} + \beta_2 X_{2k} + \beta_3 Z_{11k} + \theta_1 \beta_3 Z_{12k} + \beta_4 Z_{21k} \\ &+ \theta_2 \beta_4 Z_{22k} + \beta_5 Z_{11k} * Z_{21k} + \theta_2 \beta_5 Z_{11k} * Z_{22k} \\ &+ \theta_1 \beta_5 Z_{12k} * Z_{21k} + \theta_1 \theta_2 \beta_5 Z_{12k} * Z_{22k}\end{aligned}$$

Model II′

$$\begin{aligned}\text{logit Pr}(Y_k = 1) =& \beta_0 + \beta_1 X_{1k} + \beta_2 X_{2k} + \beta_3 E_k + \beta_4 Z_{11k} + \theta_1 \beta_4 Z_{12k} \\ &+ \beta_5 Z_{11k} * E_k + \theta_1 \beta_5 Z_{12k} * E_k\end{aligned}$$

where for $i = 1, 2$,

$$(Z_{i1k}, Z_{i2k}) = \begin{cases} (0,0) & \textit{when } G_{ik} = 0, \\ (0,1) & \textit{when } G_{ik} = \theta_i, \\ (1,0) & \textit{when } G_{ik} = 1. \end{cases}$$

Score Test

- Joint effect $H_{01} : \beta_3 = \beta_4 = \beta_5 = 0$
- Interactive effect $H_{02} : \beta_5 = 0$

Let $\Theta = (\beta_0, \beta_1, \beta_2, \beta_3, \beta_4, \beta_5)^\tau = (\phi, \boldsymbol{\eta})^\tau$. Denote ϕ being the interested parameters, and $\boldsymbol{\eta}$ as the nuisance parameters.

Note: Under H_{01}, the parameters θ is unestimable from the data. The traditional tests, such as score test or likelihood-ratio test, which need to estimate all the nuisance parameters under the null hypothesis, are improper.

Score Test

We fix the values of θ, a score statistic can be formed for testing H_{0u}. Varying the values of θ from a appropriate range, we can obtain a family of score statistics.

We propose to use the maximum value of such score statistics as the final test statistics.

Score Test

Denote the maximum likelihood estimation (MLE) as $\hat{\Theta}_u = (0, \hat{\boldsymbol{\eta}}_u)^\tau$, which maximizes $l(\Theta_u; \theta)$ under the null hypothesis H_{0u}.

Assume that $p_k = \Pr(Y_k = 1)$, for fixed value of θ, the logarithm Likelihood function for testing H_{0u} is

$$l(\Theta_u; \theta) = \sum_{k=1}^{n} \left[Y_k \log p_k + (1 - Y_k) \log(1 - p_k) \right].$$

Score Test

The score function can be written as

$$U(\Theta_u;\theta) = \begin{bmatrix} U_{\phi_u}(\Theta_u;\theta) \\ U_{\boldsymbol{\eta}_u}(\Theta_u;\theta) \end{bmatrix} = \begin{bmatrix} \frac{\partial l(\Theta_u;\theta)}{\partial \phi} \\ \frac{\partial l(\Theta_u;\theta)}{\partial \boldsymbol{\eta}} \end{bmatrix}.$$

The observed Fisher information matrix $\mathrm{I}_n(\Theta_u;\theta)$ can be decomposed as

$$\mathrm{I}_n(\Theta_u;\theta) = -\frac{\partial^2 l(\Theta_u;\theta)}{\partial\Theta_u\partial\Theta_u^\tau} = \begin{bmatrix} i_{\phi_u\phi_u}(\Theta_u;\theta) & i_{\phi_u\boldsymbol{\eta}_u}(\Theta_u;\theta) \\ i_{\boldsymbol{\eta}_u\phi_u}(\Theta_u;\theta) & i_{\boldsymbol{\eta}_u\boldsymbol{\eta}_u}(\Theta_u;\theta) \end{bmatrix}.$$

The inverse of $\mathrm{I}_n(\Theta_u;\theta)$ is

$$\mathrm{I}_n^{-1}(\Theta_u;\theta) = \begin{bmatrix} i^{\phi_u\phi_u}(\Theta_u;\theta) & i^{\phi_u\boldsymbol{\eta}_u}(\Theta_u;\theta) \\ i^{\boldsymbol{\eta}_u\phi_u}(\Theta_u;\theta) & i^{\boldsymbol{\eta}_u\boldsymbol{\eta}_u}(\Theta_u;\theta) \end{bmatrix}.$$

Score Test

The proposed test can be computed as

$$\text{MEST} = \max_{0\le\theta\le 1} T(\hat{\Theta}_u;\theta) = \max_{0\le\theta\le 1}\left\{U_{\phi_u}^{\tau}(\hat{\Theta}_u;\theta)\mathrm{I}^{\phi_u\phi_u}(\hat{\Theta}_u;\theta)U_{\phi_u}(\hat{\Theta}_u;\theta)\right\}.$$

Specifically, the corresponding MEST for Model I′ and Model II′ are $\max_{0\le\theta_1\le 1, 0\le\theta_2\le 1} T(\hat{\Theta}_u;\theta_1,\theta_2)$ and $\max_{0\le\theta_1\le 1} T(\hat{\Theta}_u;\theta_1)$, respectively.

Algorithm

How to vary the values of θ, and then compute the ST?

- Model I$'$: scatter many points equidistantly in the interval $[0, 1]$
- Model II$'$: a sequential algorithm (SNTO) (Fang and Wang, 1993)

SNTO Algorithm

Step 0 **Initialization.** Set $t = 0, D^{(0)} = D, a^{(0)} = (0,0)$, and $b^{(0)} = (1,1)$.
Step 1 **Generate an NT-net.** Use a number-theoretic method to generate n_t points $P^{(t)}$ uniformly scattered on $D^{(t)} = [a^{(t)}, b^{(t)}]$.
Step 2 **Compute a new approximation.** Find $\boldsymbol{\theta}^{(t)} \in P^{(t)} \cup \{\boldsymbol{\theta}^{(t-1)}\}$ and $M^{(t)}$ such that

$$M^{(t)} = \mathrm{MEST}(\boldsymbol{\theta}^{(t)}) \geq \mathrm{MEST}(\boldsymbol{\psi}), \quad \forall \boldsymbol{\psi} \in P^{(t)} \cup \{\boldsymbol{\theta}^{(t-1)}\},$$

where $\boldsymbol{\theta}^{(-1)}$ is the empty set.

SNTO Algorithm

Step 3 **Termination criterion.** Let $c^{(t)} = (b^{(t)} - a^{(t)})/2$. If $\max\{c^{(t)}\} < \delta$, a pre-assigned small number, then $\boldsymbol{\theta}^{(t)}, M^{(t)}$ are acceptable; terminate algorithm. Otherwise, proceed to next step.

Step 4 **Contract domain.** Form new domain $D^{(t+1)} = [a^{(t+1)}, b^{(t+1)}]$ as follows, for $i = 1, 2$:

$$a_i^{(t+1)} = \max(\theta_i^{(t)} - \gamma c_i^{(t)}, a_i)$$

and

$$b_i^{(t+1)} = \min(\theta_i^{(t)} + \gamma c_i^{(t)}, b_i)$$

where γ is a predefined contraction ratio. Set $t = t + 1$. Go to Step 1.

Simulation Settings

We mainly compare with the AST, the CML (Chatterjee and Carroll, 2005), the EBM (Mukherjee et al., 2008).

Parameter settings

- Model I′, H_{01}:
  $\beta_0 = -2.94, \beta_1 = 0.8, \beta_2 = 1, \beta_3 = \log(1.2), \beta_4 = \log(1.2), \beta_5 = \log(1.4)$
- Model I′, H_{02}:
  $\beta_0 = -2.94, \beta_1 = 0.8, \beta_2 = 1, \beta_3 = \log(1.6), \beta_4 = \log(1.6), \beta_5 = \log(1.8)$
- Model II′, H_{01}:
  $\beta_0 = -2.94, \beta_1 = 0.8, \beta_2 = 1, \beta_3 = \log(1.2), \beta_4 = \log(1.2), \beta_5 = \log(1.4)$
- Model II′, H_{02}:
  $\beta_0 = -2.94, \beta_1 = 0.8, \beta_2 = 1, \beta_3 = \log(1.4), \beta_4 = \log(1.4), \beta_5 = \log(1.6)$

Simulation Settings

Sample Size

- Model I′, joint effect: 1400 cases and 1400 controls
- Model I′, interactive effect: 2500 cases and 2500 controls
- Model II′, joint effect: 800 cases and 800 controls
- Model II′, interactive effect: 2400 cases and 2400 controls

Variables

- $(X_1, X_2)^\tau \sim N(\boldsymbol{\mu}, \mathrm{I})$, where $\boldsymbol{\mu} = (1.2, 1.4)^\tau$.
- G_1, G_2 are sampled from trinomial distributions under the HWE.
- E are sampled from binomial distribution $Binom(0.3)$ followed by Han et al.(2012), and $N(0, 0.5)$.

The Empirical Type I Error Rates

Table 1. The empirical type I error rates of the MEST, the AST, the CML, and the EBM for Model I′.

MAF	0.05	0.15	0.25	0.35	0.45
			joint effect		
MEST	0.048	0.035	0.047	0.042	0.054
AST	0.044	0.04	0.048	0.041	0.048
CML	0.043	0.036	0.061	0.043	0.049
EBM	0.035	0.027	0.043	0.032	0.03
MAF	0.05	0.15	0.25	0.35	0.45
			interactive effect		
MEST	0.041	0.044	0.045	0.043	0.036
AST	0.034	0.051	0.063	0.052	0.057
CML	0.054	0.057	0.06	0.067	0.047
EBM	0.031	0.031	0.027	0.043	0.026

The Empirical Type I Error Rates

Table 2. The empirical type I error rates of the MEST, the AST, the CML, and the EBM for Model II′, E are generated from $Binom(0.3)$.

MAF	0.05	0.15	0.25	0.35	0.45
			joint effect		
MEST	0.046	0.056	0.043	0.064	0.058
AST	0.055	0.055	0.045	0.055	0.048
CML	0.054	0.053	0.047	0.046	0.058
EBM	0.031	0.038	0.032	0.028	0.04
MAF	0.05	0.15	0.25	0.35	0.45
			interactive effect		
MEST	0.063	0.057	0.052	0.05	0.047
AST	0.059	0.067	0.043	0.051	0.046
CML	0.051	0.064	0.057	0.048	0.044
EBM	0.047	0.047	0.045	0.04	0.037

The Empirical Power

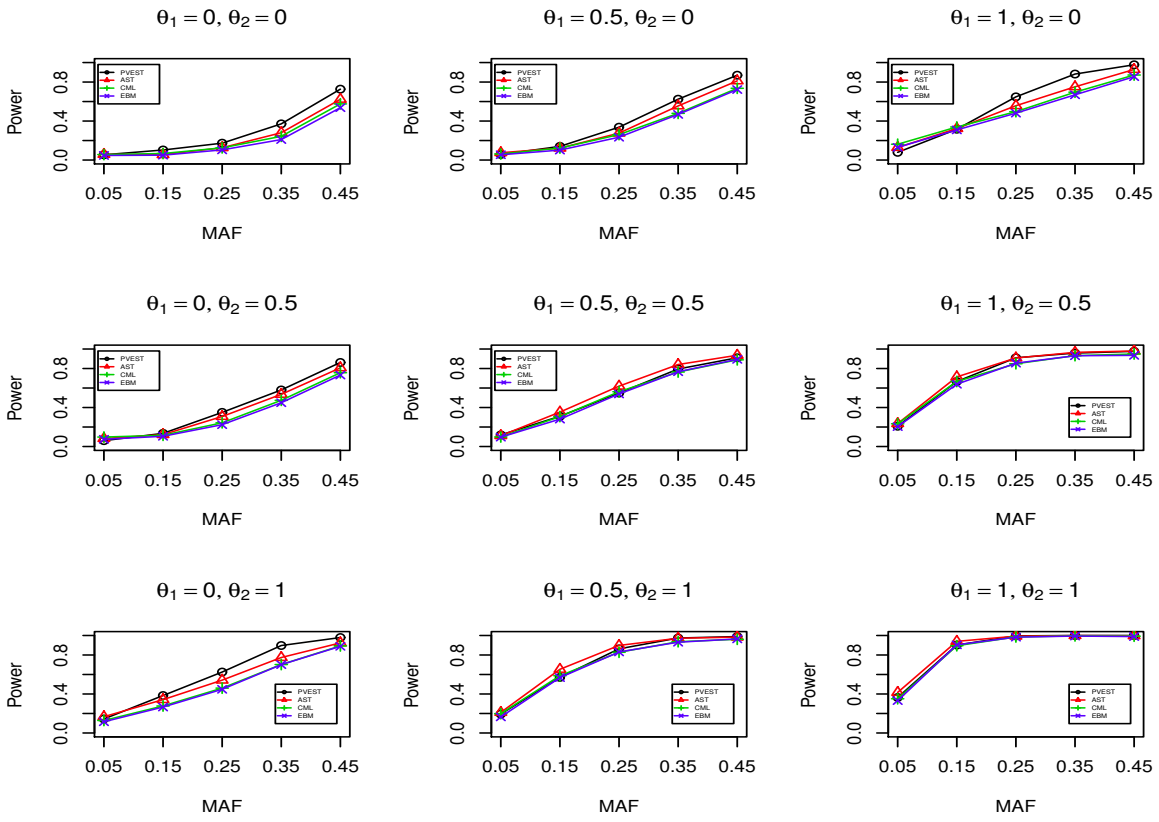

Figure: The empirical power of the MEST, the AST, the CML, and the EBM for Model I′ to test joint effect under various genetic models based on 1,000 replicates.

The Empirical Power

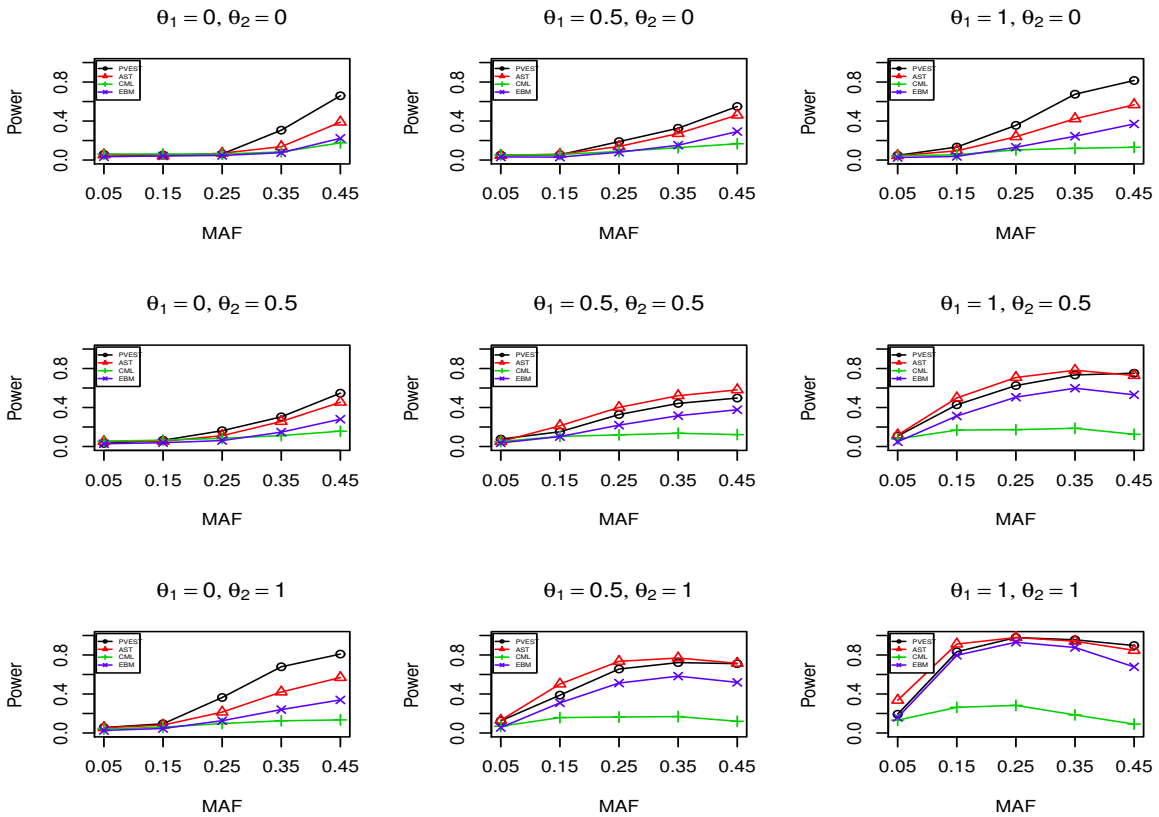

Figure: The empirical power of the MEST, the AST, the CML, and the EBM for Model I′ to test interactive effect under various genetic models based on 1,000 replicates.

The Empirical Power

Table 3. The empirical power of the MEST, the AST, the CML, and the EBM for Model II′ under three genetic models. E are generated from $Binom(0.3)$.

MAF	0.05	0.15	0.25	0.35	0.45
		$\theta_1 = 0$, interactive effect			
MEST	0.054	0.096	0.232	0.39	0.69
AST	0.049	0.075	0.16	0.311	0.549
CML	0.049	0.068	0.095	0.141	0.228
EBM	0.03	0.043	0.089	0.182	0.369

The Empirical Power

Table 3. The empirical power of the MEST, the AST, the CML, and the EBM for Model II′ under three genetic models. E are generated from $Binom(0.3)$.

MAF	0.05	0.15	0.25	0.35	0.45
		$\theta_1 = 0.5$, interactive effect			
MEST	0.113	0.344	0.463	0.582	0.556
AST	0.16	0.377	0.522	0.598	0.634
CML	0.1	0.173	0.188	0.231	0.217
EBM	0.092	0.222	0.321	0.391	0.433

The Empirical Power

Table 3. The empirical power of the MEST, the AST, the CML, and the EBM for Model II′ under three genetic models. E are generated from $Binom(0.3)$.

MAF	0.05	0.15	0.25	0.35	0.45
		$\theta_1 = 1$, interactive effect			
MEST	0.455	0.797	0.857	0.855	0.786
AST	0.48	0.814	0.855	0.809	0.725
CML	0.189	0.307	0.296	0.274	0.202
EBM	0.281	0.621	0.688	0.62	0.523

THANKS!